CryptAlyst Futures — a dedicated platform

Reusing ATLAS + the envelope; rebuilding everything above the bus

> Architectural plan for a **futures and commodities** trading platform — energy, metals, agriculture, rates — built on the same operating system as the crypto platform (the envelope, the bus, the decoupled model→signal→book→strategy), but with a new futures domain on top: instruments, forward curves, margin, FIX, compliance, and fundamental modelling.

---

1. The advice, stated plainly

Do not extend the crypto platform. Build a new platform that reuses its core.

The envelope and ATLAS are *market-agnostic*. A packet is a packet — the bus does not know or care whether it carries a BTCUSDT depth tick or a Henry Hub natural-gas forward curve. That is precisely why the core carries over unchanged.

What must be new is **everything above the bus**: the instruments, the data, the models, the execution, the risk and the compliance. Futures are not crypto, and the differences are structural:

Crypto (spot)	Futures
single asset, no expiry	an **underlying** + an **expiry** + a **roll**
one spot price	a **forward curve** and **basis**
perpetual funding	**margin** (initial/variation) and **margin calls**
unregulated, un-cleared	**cleared**, **FIX**, **MiFID II / EMIR**

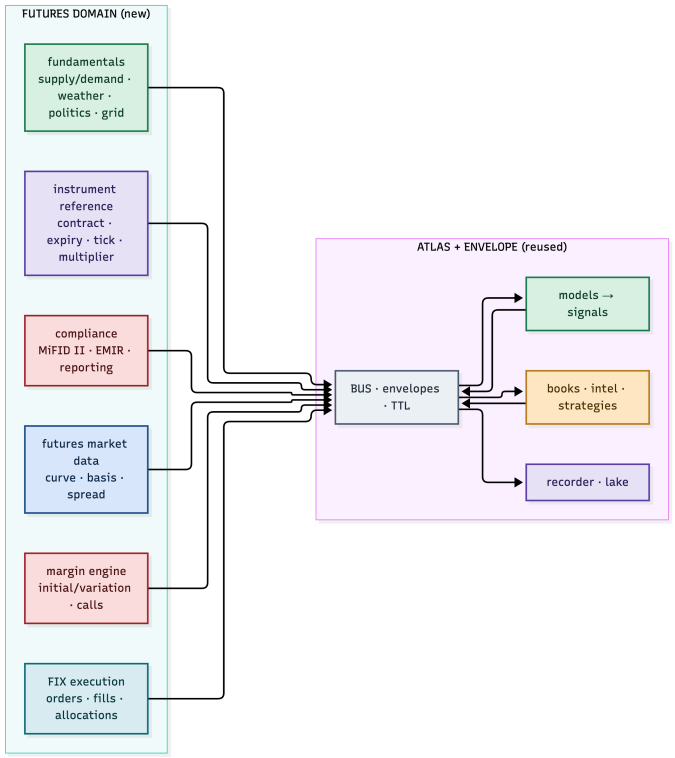


Figure 1.1 — the envelope/bus/model/book/lake core is reused; the futures domain is built new.

---

2. The planes — the futures domain

The crypto platform has five planes. The futures platform adds or reshapes them:

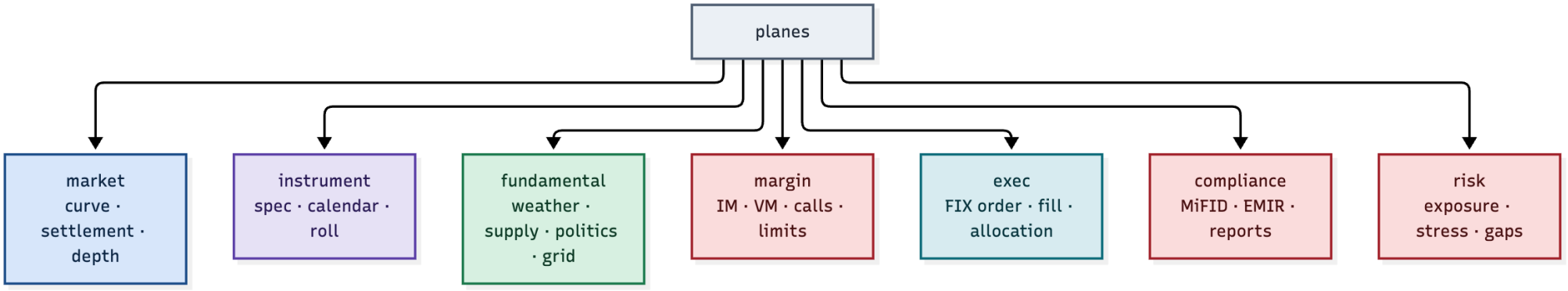


Figure 2.1 — market, instrument, fundamental, margin, exec, compliance, risk.

Plane	What it carries	New or adapted
`market`	curve, settlement, depth, spread	adapted (curve, not spot)
`instrument`	contract spec, calendar, roll	**new**
`fundamental`	supply/demand, weather, politics, grid	**new**
`margin`	IM, VM, calls, limits	**new**
`exec`	FIX order, fill, allocation	**new**
`compliance`	MiFID II, EMIR, reports	**new**
`risk`	exposure, stress, gap	adapted

Every one of these is a *producer of envelopes* on the same bus. The model, book and strategy layers subscribe exactly as they do today — they just consume futures data.

---

3. Instrument reference data

Futures cannot be traded without knowing *what a contract is*. The instrument domain is the single source of truth for every contract:

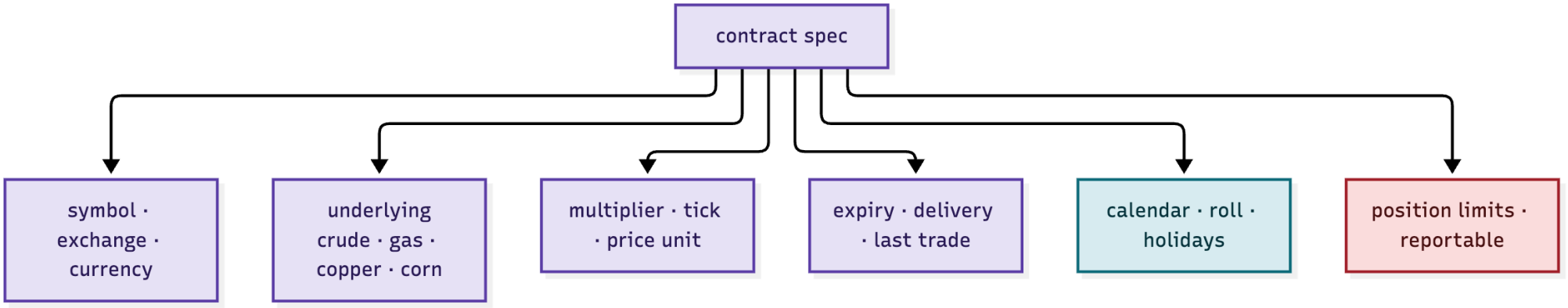


Figure 3.1 — symbol, underlying, multiplier, tick, expiry, calendar, limits.

Field	Example	Why it matters
symbol / exchange	CL (CME), TTF (ICE), F (EEX)	the address of the contract
underlying	WTI crude, Dutch gas, copper	what you are actually exposed to
multiplier / tick	1,000 bbl × $0.01	the P&L per tick
expiry / delivery	M1 … M12, last trade date	when the contract rolls
calendar / roll	roll dates, holidays	how the curve is built and rolled
limits	position, reportable	regulatory + exchange caps

The instrument domain is the *reference data problem* in miniature — and in a futures platform it is the foundation every other component keys off.

---

4. Market data — the forward curve

Futures price is not a single number. It is a **curve**: one price per delivery month. And the shape of that curve is itself the signal.

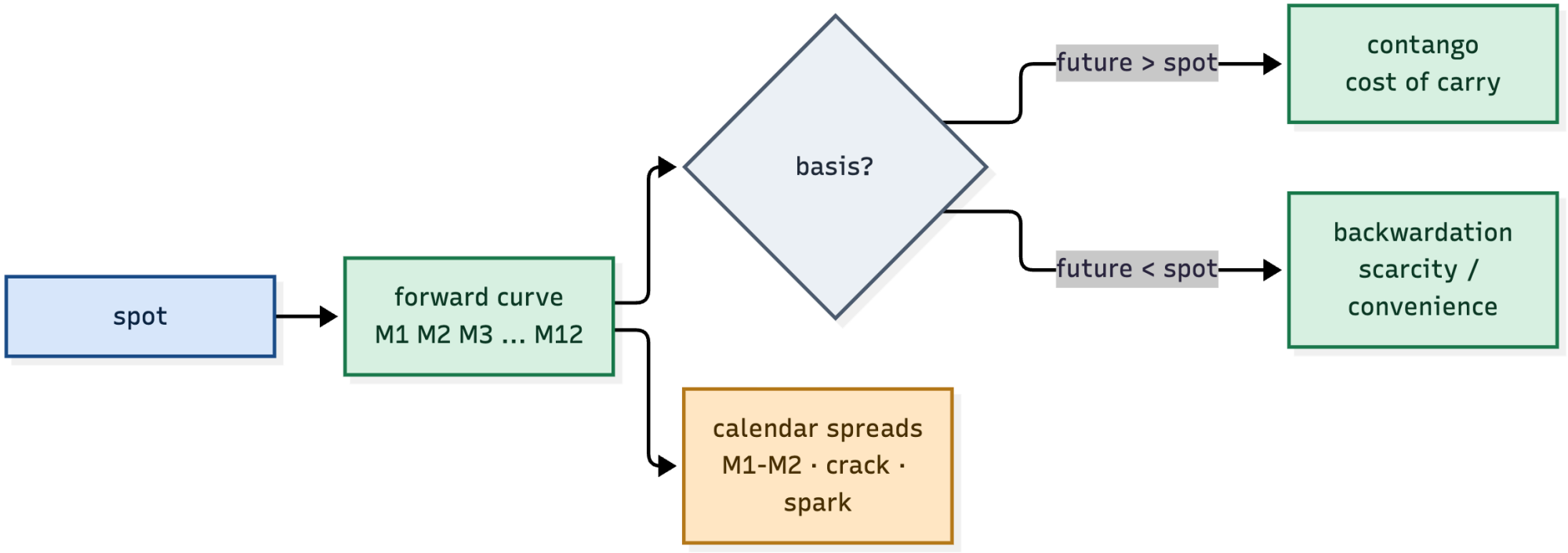


Figure 4.1 — contango and backwardation are the market's opinion of carry and scarcity.

Term	Meaning
contango	futures above spot — cost of carry (storage, interest)
backwardation	futures below spot — scarcity or convenience yield
basis	future – spot
calendar spread	the price of one month vs the next (M1–M2)
crack / spark	refining / power-generation margins

To *work out the price — flat or forward* — you must understand carry, storage, seasonality and the shape of the curve. That is the difference from crypto, and it is the whole game.

5. Margin — the risk you cannot ignore

Trading futures **bears risk**: you post initial margin, you are marked to market daily (variation margin), and when the account falls below maintenance you get a **margin call** — top up or be liquidated.

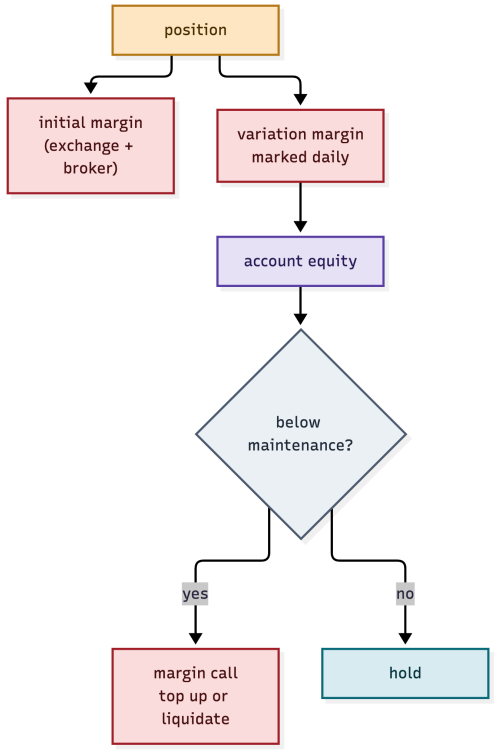


Figure 5.1 — initial margin, variation margin, and the margin call path.

Margin	What it is
initial margin	posted to open a position
variation margin	the daily mark-to-market cash flow
maintenance	the floor before a call
margin call	the demand to top up or be closed

The margin engine is a **first-class producer on the bus**: it publishes `margin.state` and `margin.call` envelopes, and the risk governor refuses any order that would breach free margin. In futures, the margin check happens *before* the order — not after.

---

6. Execution — FIX

Futures orders go to regulated venues (CME, ICE, EEX, LME) through brokers — and that is **FIX**. The exec plane speaks FIX 4.4:

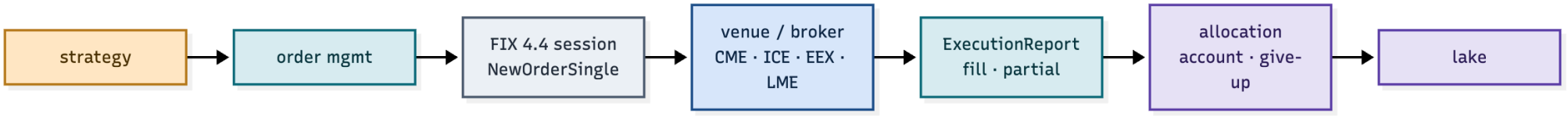


Figure 6.1 — NewOrderSingle → venue → ExecutionReport → allocation → lake.

FIX message	Purpose
`NewOrderSingle` (D)	send an order
`ExecutionReport` (8)	acknowledge, partial, fill
`OrderCancelRequest` (F)	cancel
allocation	assign a fill to accounts (give-up)

The FIX session is *another envelope producer/consumer*: every order, fill and allocation is a stamped packet on the bus, correlated to the strategy that generated it — so the audit trail from signal to fill is complete, exactly as in crypto.

---

7. Compliance — MiFID II and EMIR

Regulation is **not optional**. The futures platform must satisfy, at minimum:

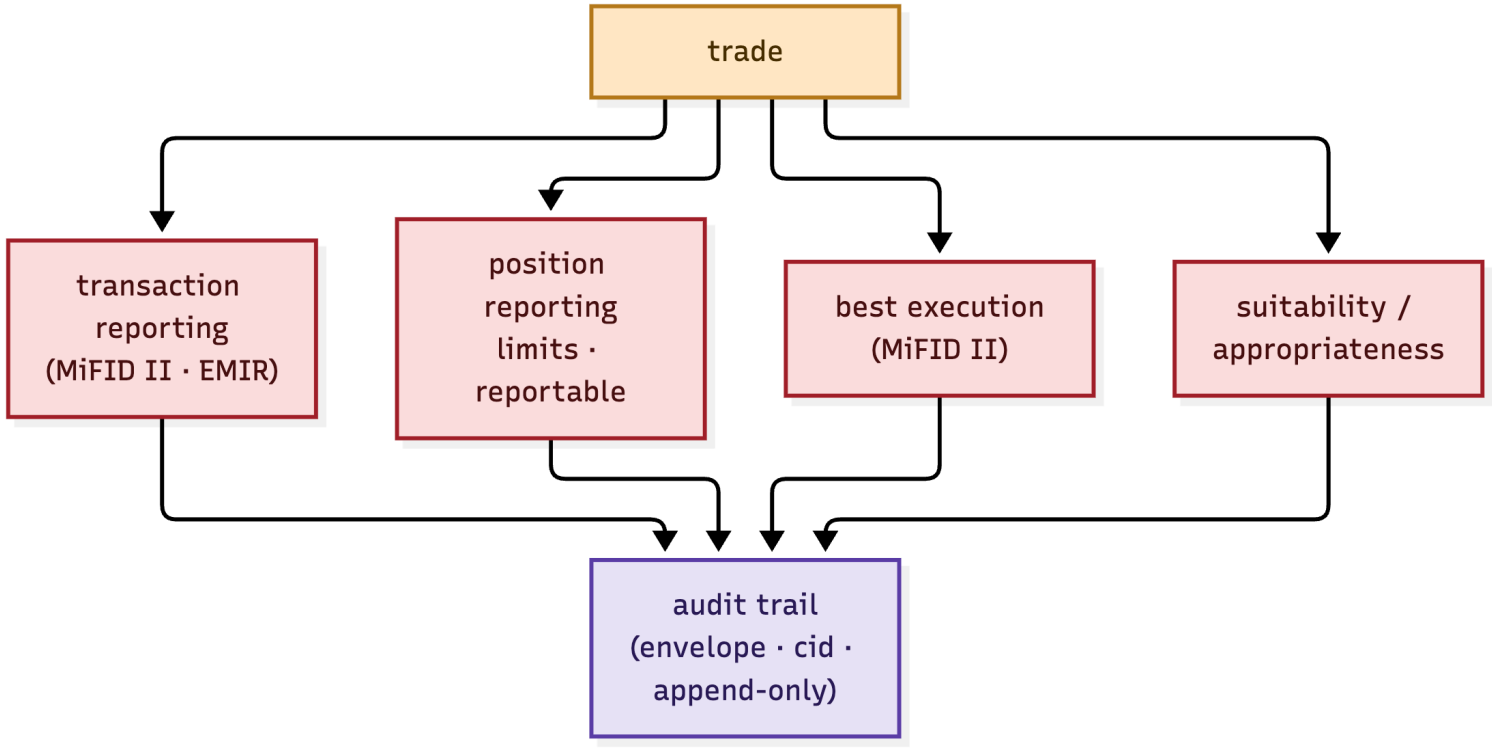


Figure 7.1 — transaction reporting, position reporting, best execution, suitability.

Obligation	Regime	What it requires
transaction reporting	MiFID II / EMIR	report every trade, on time
position reporting	MiFID II / EMIR	report holdings past limits
best execution	MiFID II	prove you got the best price
suitability	MiFID II	know your client and their risk

Because the envelope carries a correlation id and the lake is append-only, the compliance report is **generated from the same evidence** as the trading record — a report is a view over the lake, not a separately-maintained log. That is the envelope "having you covered".

8. Fundamentals — weather, politics, the grid

The crypto models read order flow. The futures models must also read the *world*: supply and demand, weather, politics, and — for energy — the grid itself.

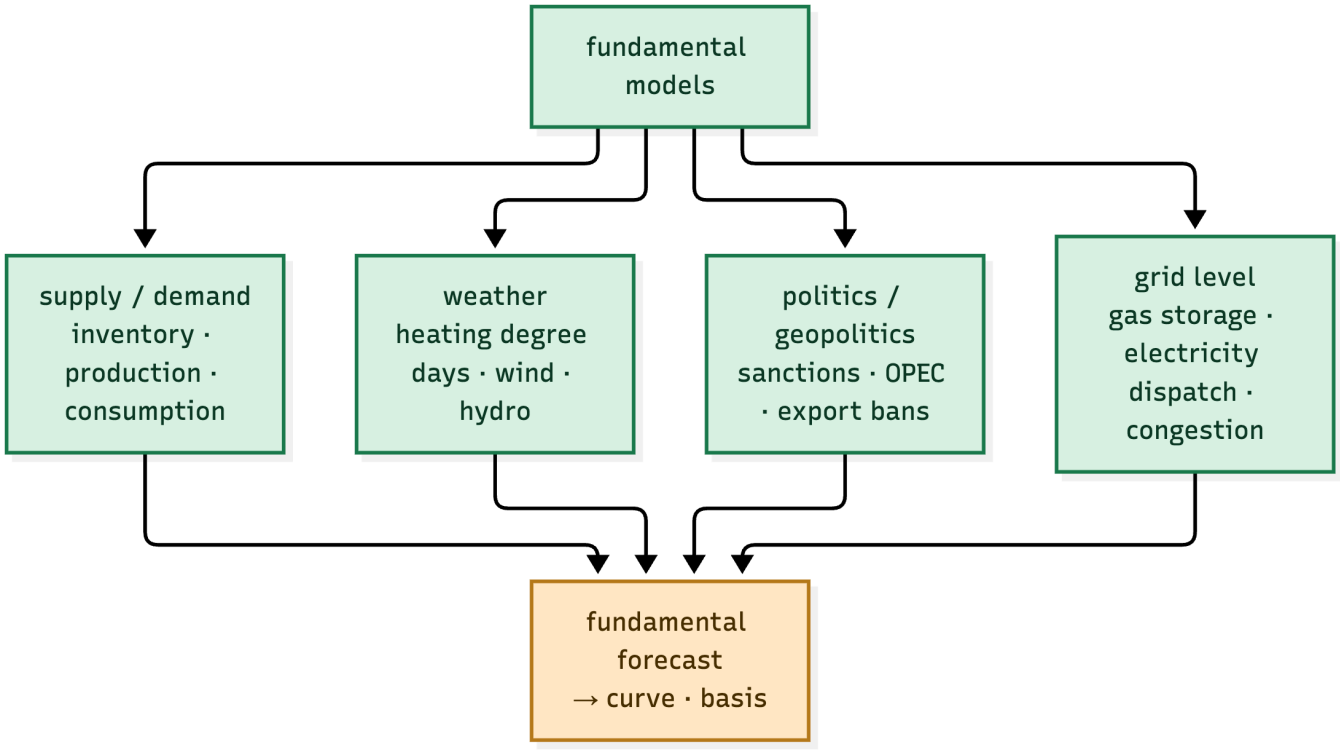


Figure 8.1 — supply/demand, weather, politics and grid feed the fundamental forecast.

Fundamental	Drives	Example
supply / demand	inventory, production, consumption	EIA crude stocks, USDA WASDE
weather	heating/cooling demand, wind, hydro	HDD/CDD, wind forecast
politics	sanctions, OPEC, export bans	a pipeline sanction, an export tax
grid	dispatch, congestion, storage	gas storage %, power plant availability

Energy, especially, is a **systems problem**: gas and electricity prices are set at the margin by the most expensive plant, on the most congested node, at the most extreme weather. Modelling that — down to the grid level — is a first-class model domain, publishing its forecast as an envelope the strategies consume.

9. Risk — the governor, futures-grade

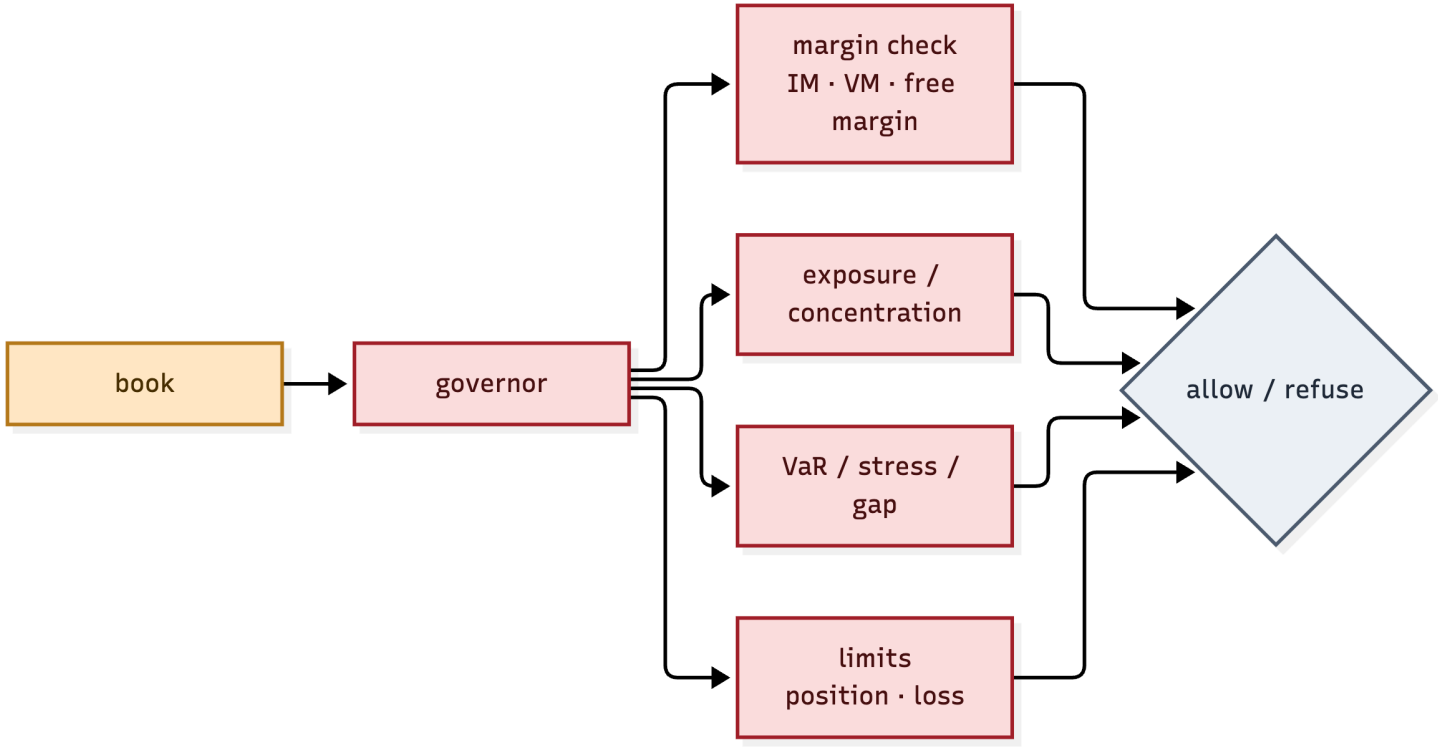


Figure 9.1 — margin, exposure, VaR/stress/gap and limits, all before an order.

Check	Futures-specific
margin	free margin ≥ IM for the order
exposure	net and gross, per underlying and per curve bucket
stress / gap	what happens at expiry, or on a limit move
limits	exchange position limits, reportable thresholds

The governor is the same idea as crypto — refuse with a rule and a number — but the rules are futures-grade: margin, concentration, and gap risk on delivery.

---

10. The models

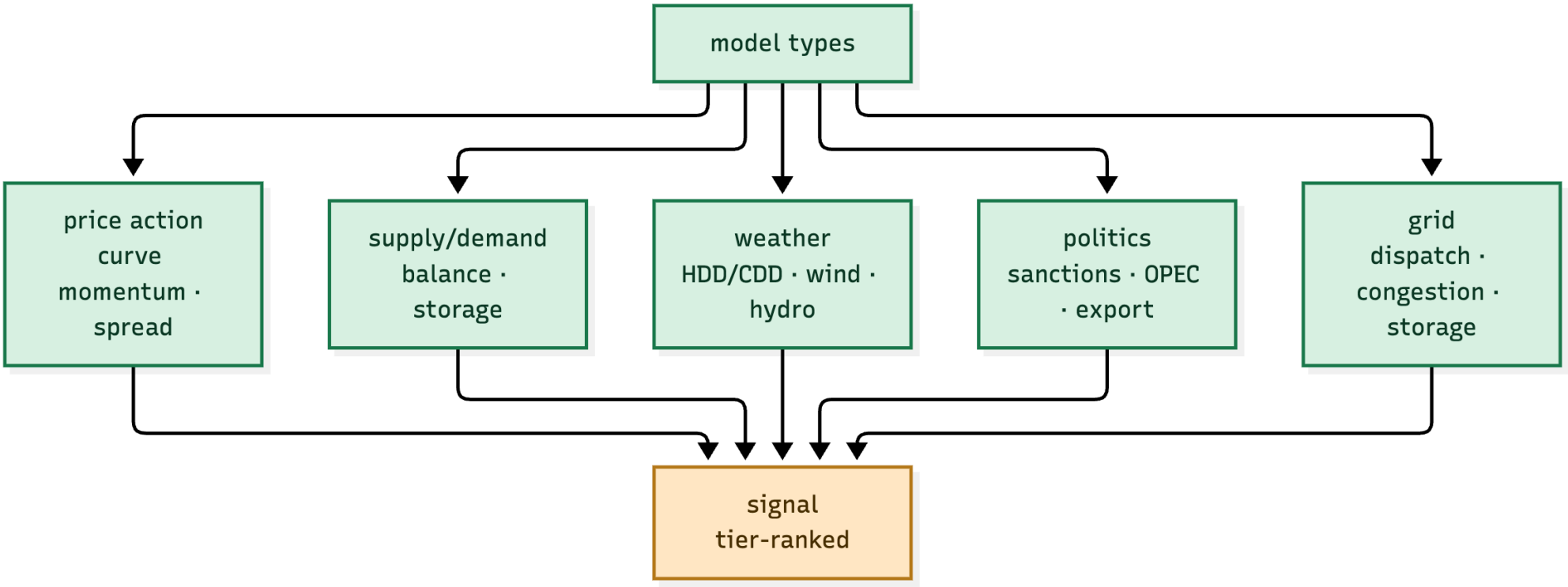


Figure 10.1 — price action, supply/demand, weather, politics, grid — each a pure signal producer.

Model	Question
price action	is the curve trending, is the spread moving?
supply/demand	is the balance tightening or loosening?
weather	is demand or renewable supply shifting?
politics	is a supply shock forming?
grid	is the system at a stress point?

Each is a **calculation agent** — a pure function of its inputs, publishing a signal with an epoch timestamp. The book accumulates them in its intel window and confirms a pattern before a strategy trades. Advisory, systematic and auto are just three *strategies* over the same signals.

11. Crypto vs futures, side by side

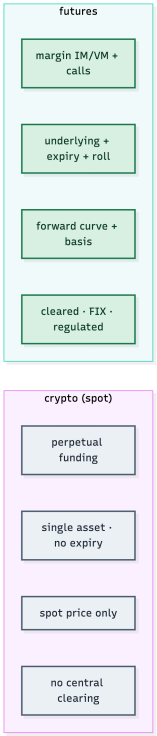


Figure 11.1 — spot vs forward: expiry, curve, margin, clearing.

The one line that matters: **crypto trades an asset; futures trade a curve of commitments** — and everything new in this platform (instrument, margin, FIX, compliance, fundamentals) exists because of that difference.

---

12. Build plan — phases

Phase	What	Depends on
1 · Core reuse	fork the envelope/bus/model/book/lake core	—
2 · Instrument + curve	reference data, forward curve, basis	1
3 · Market data	venue feeds (CME, ICE, EEX, LME)	1, 2
4 · Margin + risk	margin engine, governor	2, 3
5 · Execution	FIX 4.4 session, order/fill/allocation	3, 4
6 · Compliance	MiFID II / EMIR reporting	5
7 · Fundamentals	supply/demand, weather, politics, grid	2
8 · Strategies	advisory → systematic → auto	4, 7

Each phase ships envelopes on the same bus, so the platform is *observable and auditable at every step* — which is the entire point of ATLAS and the envelope.

---

13. Summary

The futures platform is **the same operating system, a different market**. Reuse the envelope, the bus, the decoupled model→signal→book→strategy, the intel window and the lake. Build new: instruments, forward curves, margin, FIX, compliance and fundamental modelling. The envelope covers the *how*; the futures domain is the *what*.